

Chemistry mix v2

Chemistry (4CH1) · IGCSE · total 30 marks · name: _____

Question 1 (1 mark)

(a)

[1]

Which diagram shows the electronic configuration of a nitrogen atom?

.....

.....

.....

Question 2 (1 mark)

(a)

[1]

The following elements are found in group 0 of the periodic table.

He Ne Ar Kr Xe Rn

What explains their similar chemical and physical properties?

- A) They have the same number of valence electrons
- B) They are all gases
- C) They have the same number of protons and electrons
- D) They have full outer shells

Question 3 (1 mark)

(a)

[1]

Which element, T, Q, R, or M has the electronic configuration 2,8,3?

																M	
	Q											R					
T																	

- A) element M
- B) element Q
- C) element T
- D) element R

Question 4 (4 marks)

(a) give

[1]

The diagram shows part of the Periodic Table, with elements represented by the letters L, M, Q, R and T.

The letters in the diagram represent elements but are not their chemical symbols.

1 2		3 4 5 6 7 0					
L							
M							

Give the letter from the diagram that represents a noble gas.

(b) state

[1]

Elements L and M are in the same group.

State why they have similar chemical reactions.

(c) explain

[2]

An atom of element Q has 31 protons.

Use this information to explain how you can determine the number of protons in an atom of element R.

Question 5 (4 marks)

(a) complete

[4]

The table contains statements about elements in the Periodic Table.

Complete the missing information.

[4]

Statement
The name of the element with atomic number 18.
The name of the element with relative atomic mass 7.
The name of the element in Group 3, Period 4.
The electronic configuration of the element phosphorus.

Question 6 (5 marks)

(a)

[1]

In 1937, an airship full of hydrogen gas flew from Germany to America.



(Source: © Michael Rosskothen/Shutterstock)

Which property of hydrogen makes it a suitable gas to use in an airship?

- A) colourless
- B) insoluble in water
- C) low density
- D) no smell

(b) explain

[2]

Explain why helium is now used in airships instead of hydrogen.

.....

.....

.....

(c) multiple

[2]

Hydrogen is used to manufacture ammonia, NH_3

Hydrogen is reacted with nitrogen using an iron catalyst.

i) Give a chemical equation for this reaction.

(1)

ii) State why a catalyst is used in this reaction.

(1)

.....

.....

.....

Question 7 (5 marks)

(a) give

[4]

The diagram shows the positions of some elements in the Periodic Table.

1	2												3	4	5	6	7	0
																		He
																	F	
Na																	Cl	
K																	Br	

Use symbols from this table to answer these questions.

Each symbol may be used once, more than once or not at all.

i) Give the symbol of a metal.

(1)

ii) Give the symbol of a noble gas.

(1)

iii) Give the symbol of a liquid at room temperature.

(1)

iv) Give the symbols of the two elements in Period 3

(1)

..... and

.....

(b) deduce

[1]

Deduce the electronic configuration of Na.

.....

Question 8 (8 marks)

(a) what

[1]

Mendeleev arranged the elements in the Periodic Table so that elements with similar properties fell into the same vertical columns. The physical property that he used was the atomic mass of the element and the chemical property was the behaviour of the oxide and hydride formed.

What term is used to describe the vertical columns of the Periodic Table?

(b) multiple

[3]

The Group 1 elements are often called the alkali metals because of their reaction with water to form metal hydroxides and hydrogen.

i) Name two alkali metals.

(1)

ii) Name the element that is in Group 1 but cannot be classed as an alkali metal.

(1)

iii) State one observation when an alkali metal reacts with water.

(1)

(c) state

[3]

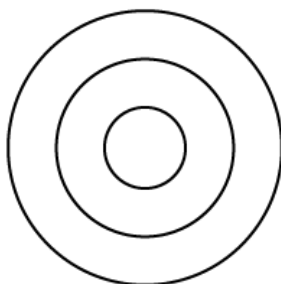
State the trends, if any, in the reactivity of elements in Groups 1, 7 and 0.

(d) complete

[1]

Argon is a Group 0 element that is found in Period 3 of the Periodic Table.

Complete the diagram to show the electronic configuration of argon.

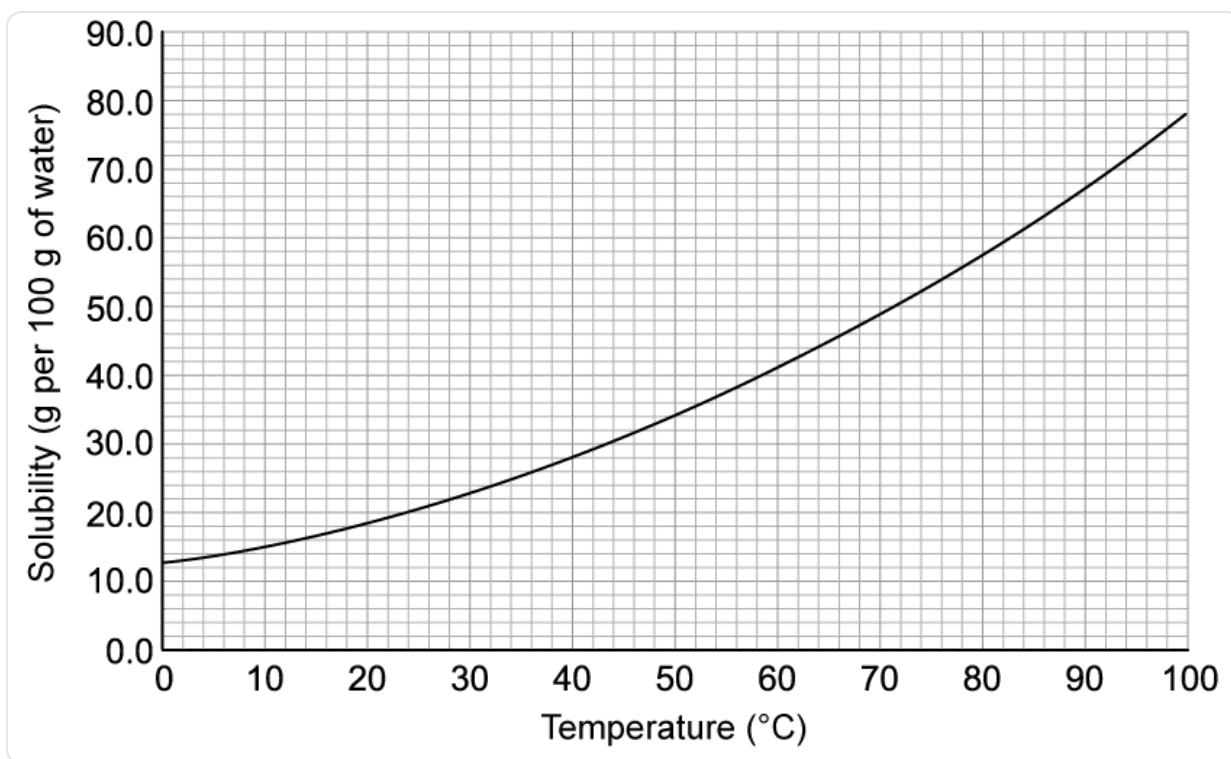


Question 9 (1 mark)

(a)

[1]

The solubility curve for copper(II)sulfate is shown below:



Which of these statements is supported by the graph?

- A) At 50 °C the solubility of copper(II) sulfate is 71 g per 100 g water
- B) At 60 °C the solubility of copper(II) sulfate is 41 g per 100 g of water
- C) The solubility of copper(II) sulfate doubles for every 10 °C rise
- D) 30 g of copper(II) sulfate can dissolve in 100 g of water at 20 °C